# Indian Banking Transactions Analysis Using Excel

Project Documentation

**Project Status:** Completed  **Tool:** Microsoft Excel

## 1. Project Overview

This project analyzes Indian banking transaction data using Microsoft Excel. The workflow covers data understanding, data cleaning and preprocessing, feature engineering, analysis, visualization, interactive dashboard development, and business insights.

## 2. Business Objectives

Analyze banking transaction patterns, understand regional and payment activity, examine branch-level fraud flags, track key transaction metrics, and present findings through an interactive dashboard.

## 3. Dataset Overview

The working dataset contains 500 transaction records and 16 columns: Transaction ID, Customer ID, Customer Name, Account Number, Region, City, Branch Name, Account Type, Transaction Type, Transaction Amount, Payment Mode, Transaction Date, Transaction Status, Balance After Txn, Is Fraud, and year_month.

## 4. Data Understanding

The dataset was reviewed to understand transaction, customer, account, geographic, payment, date, status, balance, and fraud-related fields before analysis.

## 5. Data Cleaning & Preprocessing

Duplicate records were checked using Transaction ID. Missing values were checked with COUNTBLANK and no blank cells were found. Categorical fields were reviewed for text consistency. Conditional formatting was used to check Transaction Amount and Balance After Txn for negative values; no negative values were highlighted. The observed transaction-date range was 11-Jul-2023 to 06-Jul-2025.

## 6. Feature Engineering

Analytical fields created were Year-Month, Net Change, Fraud Flag, Transaction Category, and Customer Type.

## 7. Key Performance Indicators

The completed dashboard uses four KPIs: Total Transactions, Total Transaction Amount, Average Transaction Amount, and Fraud-Flagged Transactions.

| KPI | Result |
|---|---|
| Total Transactions | 500 |
| Total Transaction Amount | ■9,39,33,780 |
| Average Transaction Amount | ■1,87,867.60 |
| Fraud-Flagged Transactions | 13 |

## 8. Dashboard Components

- Regional Transaction Volume — compares transaction counts across regions.

- Monthly Transaction Trend — shows transaction volume over the available Year-Month periods.
- Payment Mode Usage — shows the distribution across Cash, Credit Card, Debit Card, Net Banking, and UPI.
- Fraud Transactions by Branch — compares fraud-flagged and non-fraud transactions across branches.

## 9. Interactive Slicers

Region, Payment Mode, and Transaction Type slicers were added to allow interactive filtering of the dashboard.

## 10. Feature Engineering Details

Net Change formula: **=IF(I2="Withdrawal",-J2,J2)**

Fraud Flag formula: **=IF(O2="Yes",1,0)**

Customer Type formula: **=IF(COUNTIF($B$2:$B$501,B2)=1,"One-Time","Frequent")**

Transaction Category: Cash → Cash-Based; Credit Card, Debit Card, Net Banking, and UPI → Digital.

## 11. KPI Formulas

Total Transactions: **=COUNTA(Sheet1!A2:A501)**

Total Transaction Amount: **=SUM(Sheet1!J2:J501)**

Average Transaction Amount: **=AVERAGE(Sheet1!J2:J501)**

Fraud Transactions: **=COUNTIF(Sheet1!O2:O501,"Yes")**

## 12. Key Findings

- North recorded the highest regional transaction volume in the completed chart, while East recorded the lowest.
- Debit Card represented 26% of transactions; Credit Card 20%; Cash 19%; UPI 19%; Net Banking 16%.
- 13 of 500 transactions were flagged as fraud, corresponding to a 2.6% flagged-transaction rate.
- Monthly transaction volume fluctuated across the observation period, with multiple peaks and declines.

## 13. Project Workflow

Data Understanding → Data Cleaning → Feature Engineering → PivotTable Analysis → PivotCharts → KPI Cards → Interactive Slicers → Dashboard → Business Insights

## 14. Skills Demonstrated

Microsoft Excel, data cleaning, validation, Excel formulas, feature engineering, PivotTables, PivotCharts, KPI analysis, dashboard design, slicers, data visualization, and business insight generation.

## 15. Conclusion

This project demonstrates an end-to-end Excel data analysis workflow that transforms banking transaction records into analytical fields, KPIs, interactive visualizations, and business-focused insights.